

Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup

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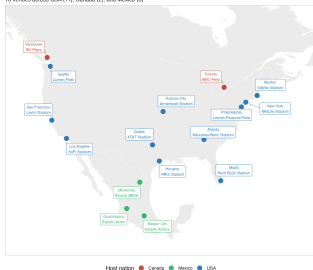
June 2026

Why does this matter?

2026 FIFA World Cup — key facts

- 48 nations, largest WC in history
- 11 US host cities, June 11 – July 19
- ~85 million international visitors projected for the US in 2026 (NTTO annual forecast)
- Visitors from every qualified nation, many in disease-endemic regions

FIFA World Cup 2026 — Venue Stadiums
16 venues across USA (11), Canada (2), and Mexico (3)



Research question

What is the probability that dengue, influenza, malaria, measles, or pertussis is imported into each US host city during June 2026 — and which source countries drive that risk?

Goal: build a quantitative importation risk baseline from open surveillance and travel data to help prioritize preparedness efforts by venue city.

Travel data

- **COR / I-94** (CBP): monthly arrivals by country of residence — June 2024 baseline
- **BTS T-100**: nonstop segment data, country-level routing fractions to each US venue city (mean June 2023–2025)
- **NTTO 2026 Forecast**: official travel projections, 12 top source markets

Disease surveillance

- **Dengue**: WHO Global Dengue Surveillance (monthly, Dec 2025)
- **Influenza**: WHO FluNet / GISRS (ISO weeks 22–26, 2023–2025)
- **Malaria**: WHO Global Malaria Programme (2024 annual)
- **Measles / Pertussis**: WHO Immunization Data portal (annual, 2025)

Open-access sources throughout — straightforward to update as new data become available.

Poisson importation framework

Core equation — expected importations of disease d into city v :

$$\Lambda_{v,d} = \sum_c \underbrace{N_{c,v}}_{\text{arrivals}} \times \underbrace{\rho_d \cdot \frac{C_{c,d}}{P_c}}_{\text{per-traveller incidence}} \times \underbrace{p_d}_{\text{travel while infectious}} \quad P(X \geq 1) = 1 - e^{-\Lambda_{v,d}}$$

Three nested models for arrivals $N_{c,v}$:

	Model	Travel volume	City routing
M1	Baseline	COR June 2024	T-100 fractions
M2	WC-adjusted	2026 NTTO projections (ϕ_c)	T-100 fractions
M3	Schedule-driven	2026 + WC fan decomposition	Match schedule

Uncertainty: ρ_d and p_d sampled from Uniform literature ranges (5,000 MC draws) $\Rightarrow$ MC median + 95% uncertainty interval on every estimate.

Three nested models — what each one adds

M1 — Baseline

No WC adjustment; routine travel only.

Growth factor:

$$\phi_c = 1 \quad \forall c$$

Arrivals:

$$N_{c,v}^{M1} = N_{c,24}^{COR} \cdot f_{c,v}^{T100}$$

$f_{c,v}^{T100}$: T-100 routing fraction (mean Jun 2023–2025)

Answers: what does baseline importation risk look like without any tournament effect?

M2 — WC-adjusted

WC travel surge applied; same routing.

Growth factor (tiered):

$$\phi_c = \begin{cases} \frac{V_{c,26}}{V_{c,24}} & \text{NTTO top-12} \\ 1.174 & \text{WC-qualified} \\ 1.077 & \text{non-qualified} \end{cases}$$

Arrivals:

$$N_{c,v}^{M2} = N_{c,24}^{COR} \cdot \phi_c \cdot f_{c,v}^{T100}$$

Answers: how much does the WC surge alone amplify risk?

M3 — Schedule-driven

Same ϕ_c as M2; fans routed by match schedule.

Travel decomposition:

$$N_c^{WC} = N_{c,24}^{COR} \cdot \max(0, \phi_c - \phi_{base})$$

$$N_c^{bg} = N_{c,24}^{COR} \cdot \phi_c - N_c^{WC}$$

Arrivals:

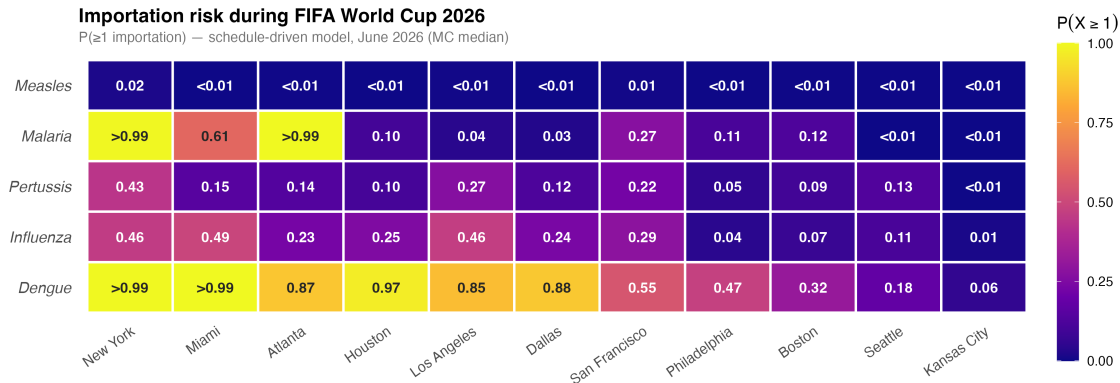
$$N_{c,v}^{M3} = N_c^{WC} \cdot \omega_{c,v} + N_c^{bg} \cdot f_{c,v}^{T100}$$

$\omega_{c,v} = g_{c,v} / G_c$: schedule routing fraction

Answers: does schedule-based fan routing shift which cities bear the highest risk?

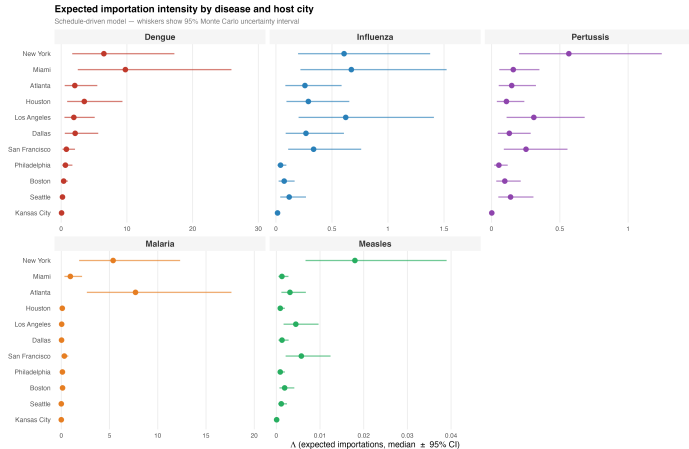
M1 → M2: isolates the WC volume effect (ϕ_c). **M2 → M3:** isolates schedule-based fan redistribution. *All results shown use M3.*

Result 1 — Importation probability across all venues



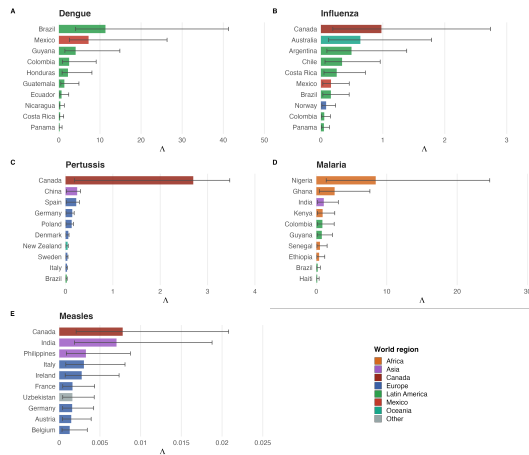
$P(\geq 1)$ under the schedule-driven model (MC median, 5,000 draws). Dengue leads at most cities ($P > 0.99$ at Miami, New York, Houston). **Atlanta exception:** malaria $>$ dengue ($P > 0.99$ vs. 0.97), driven by direct travel from West/Central Africa. Measles is consistently lowest.

Result 2 — Importation intensity with uncertainty



Bars = MC median Λ ; whiskers = 95% uncertainty interval. Dengue is orders of magnitude above all other diseases. Pertussis CIs skew downward (central value is a conservative upper bound); dengue and influenza CIs skew upward (may be understated).

Result 3 — Source country drivers



Top 10 source countries per disease (schedule-driven, all 11 US cities). Latin America dominates dengue & influenza; West/Central Africa drives malaria; Europe leads pertussis & measles. Rankings are stable across the full MC parameter range.

Conclusions

- 1 **Dengue leads importation risk** at most US venue cities ($P(\geq 1) > 0.99$ at Miami, New York, Houston) — robust across all parameter assumptions.
- 2 **Atlanta is a notable exception**: malaria probability exceeds dengue, driven by direct flights from West and Central Africa. City-level heterogeneity matters.
- 3 **Influenza ranks second** at most cities, amplified by the Southern Hemisphere winter peak during June.
- 4 **Background tourism dominates**: the WC fan increment accounts for only $\approx 8.3\%$ of projected arrivals — pre-existing travel patterns set the baseline risk.
- 5 **The framework is portable and documented**: built entirely from publicly available sources, it can be deployed for any future mass gathering with minimal lead time as travel projections and case data are updated.

A reproducible baseline intended to complement — not replace — operational intelligence and richer data streams.

Thank you

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Code & data:

`https://github.com/jdiestra1977/FIFA\_worldCup\_2026\_risk`

Preprint:

`https://www.medrxiv.org/content/10.64898/2026.06.03.26354828v1`